

## Problem set 10

### Exercise 1

In a self-attention layer, suppose the unnormalized attention score matrix is

$$S = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}.$$

Compute the softmax of  $S$  to obtain the normalized attention matrix  $A$ .

### Exercise 2

Consider a self-attention layer with the following setup:

- Sequence length:  $n = 4$
- Model dimension:  $d_{\text{model}} = 8$
- Number of heads:  $h = 2$

Assume that for each head:

$$d_k = d_v = \frac{d_{\text{model}}}{h}.$$

#### Questions:

1. What are the dimensions of  $Q$ ,  $K$ , and  $V$  for one head?
2. What is the dimension of the attention score matrix  $QK^T$ ?
3. What is the dimension of the softmax attention matrix?
4. What is the dimension of the output of one head ( $AV$ )?
5. What is the final output dimension after concatenating all heads?